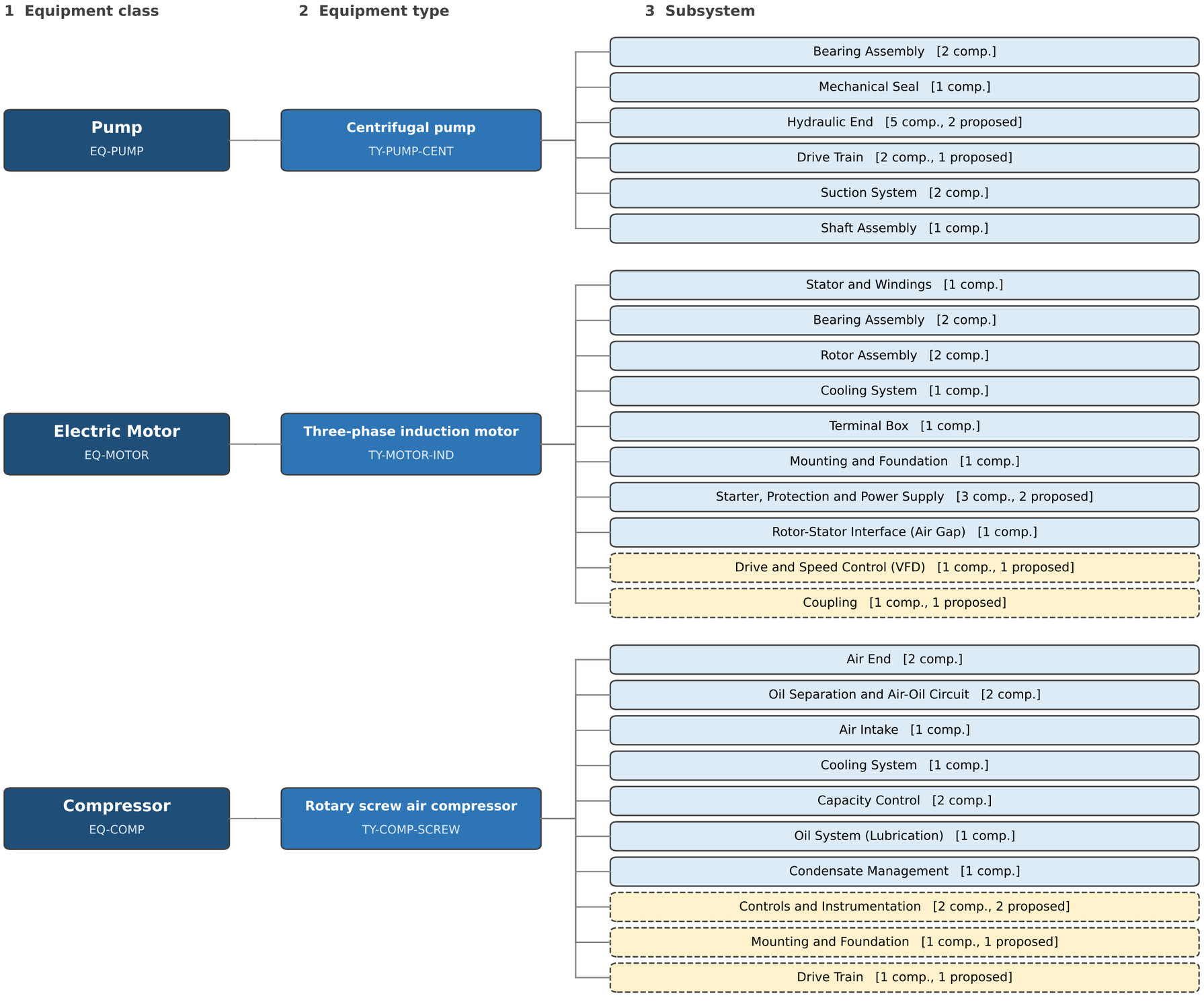
Operational Sense - Equipment Hierarchy (overview)

Level 1 class > level 2 type > level 3 subsystems. Components and failure modes are on the next pages.



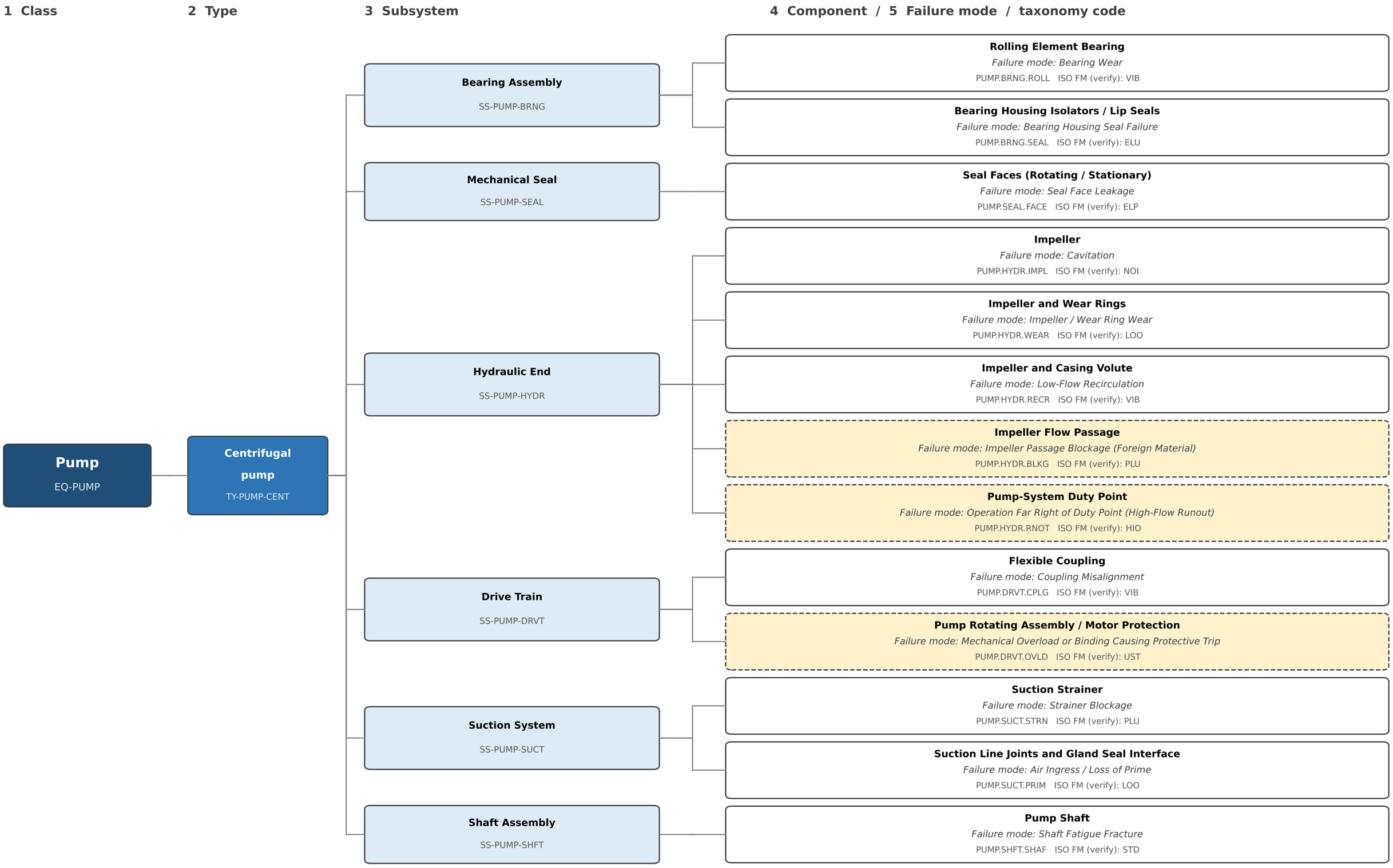
In Failure_Mode_Taxonomy

Proposed (from Taxonomy_Proposed, not approved)

Draft v0.1 - synthetic corpus, subject to M01/M02 review and Group 2/3 agreement

Pump - hierarchy detail

Centrifugal pump | 6 subsystems, 13 components. Each component box shows the failure mode and taxonomy code.



In Failure_Mode_Taxonomy

Proposed (from Taxonomy_Proposed, not approved)

Draft v0.1 - synthetic corpus, subject to M01/M02 review and Group 2/3 agreement

Electric Motor - hierarchy detail

Three-phase induction motor | 10 subsystems, 14 components. Each component box shows the failure mode and taxonomy code.

1 Class	2 Type	3 Subsystem	4 Component / 5 Failure mode / taxonomy code
Electric Motor EQ-MOTOR	**Three-phase induction motor** TY-MOTOR-IND	**Stator and Windings** SS-MOT-STAT	**Winding Insulation (Copper Magnet Wire and Varnish)** *Failure mode: Insulation Breakdown* ELEC.MOT.STAT ISO FM (verify): BRD
		Bearing Assembly SS-MOT-BRNG	**Drive-End Bearing and Grease** *Failure mode: Bearing Overheating (Lubrication Failure)* ELEC.MOT.BRNG ISO FM (verify): OHE
			Bearing (Inverter-Fed Shaft Currents) *Failure mode: Bearing Electrical Discharge Damage (Fluting)* ELEC.MOT.BRDC ISO FM (verify): VIB
		Rotor Assembly SS-MOT-ROTR	**Squirrel-Cage Rotor Bars and End Rings** *Failure mode: Broken Rotor Bar* ELEC.MOT.ROTR ISO FM (verify): VIB
			Rotor Core and Shaft *Failure mode: Rotor Imbalance* ELEC.MOT.IMBL ISO FM (verify): VIB
		Cooling System SS-MOT-COOL	**Cooling Fan and Cowl** *Failure mode: Cooling Airflow Blockage* ELEC.MOT.COOL ISO FM (verify): OHE
		Terminal Box SS-MOT-TERM	**Terminal Lugs and Connections** *Failure mode: Loose Electrical Connection* ELEC.MOT.TERM ISO FM (verify): OHE
		Mounting and Foundation SS-MOT-MNTG	**Motor Feet and Foundation Bolts** *Failure mode: Soft Foot / Mechanical Looseness* ELEC.MOT.FEET ISO FM (verify): VIB
		Starter, Protection and Power Supply SS-MOT-SUPP	**Supply Phase Conductors, Fuses and Contactor** *Failure mode: Single-Phasing (Phase Loss)* ELEC.MOT.SUPP ISO FM (verify): PDE
			Thermal Overload Relay *Failure mode: Overload Relay Trip Preventing Start* ELEC.MOT.STRT ISO FM (verify): FTS
			Overload Protection *Failure mode: Repeated Overload Trips from Excess Driven Load* ELEC.MOT.OVLD ISO FM (verify): UST
		Rotor-Stator Interface (Air Gap) SS-MOT-AGAP	**Air Gap (Rotor and Stator Core)** *Failure mode: Air-Gap Eccentricity / Rotor-Stator Rub* ELEC.MOT.AGAP ISO FM (verify): VIB
		Drive and Speed Control (VFD) SS-MOT-DRIV	**VFD Speed Feedback** *Failure mode: Speed Feedback Signal Fault (Speed Fluctuation)* ELEC.MOT.VFDF ISO FM (verify): ERO
		Coupling SS-MOT-CPLG	**Motor-Load Coupling** *Failure mode: Shaft Misalignment* ELEC.MOT.CPLG ISO FM (verify): VIB

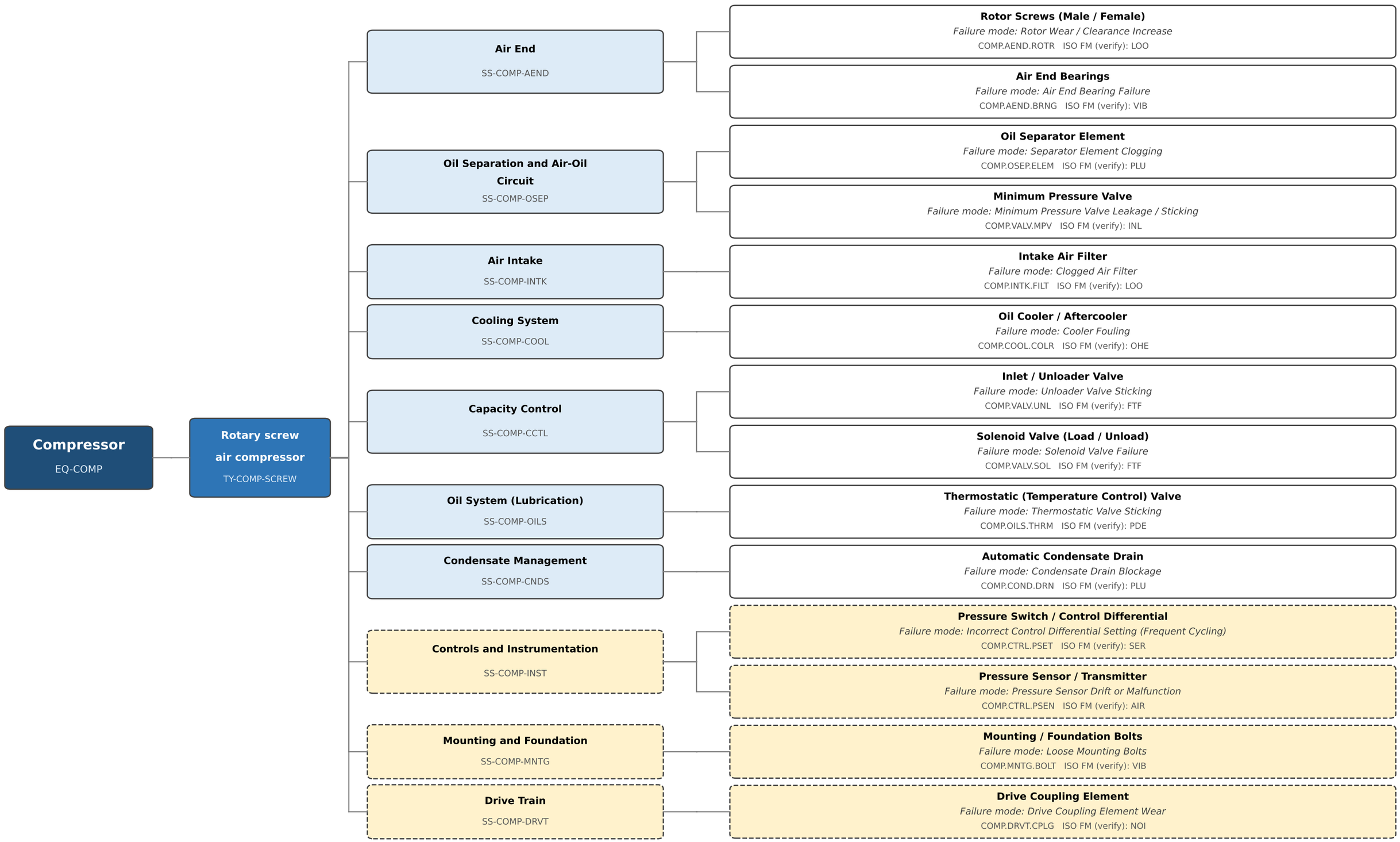
In Failure_Mode_Taxonomy

Proposed (from Taxonomy_Proposed, not approved)

Draft v0.1 - synthetic corpus, subject to M01/M02 review and Group 2/3 agreement

Compressor - hierarchy detail

Rotary screw air compressor | 10 subsystems, 14 components. Each component box shows the failure mode and taxonomy code.



In Failure_Mode_Taxonomy

Proposed (from Taxonomy_Proposed, not approved)

Draft v0.1 - synthetic corpus, subject to M01/M02 review and Group 2/3 agreement